

01 GEOMETRY

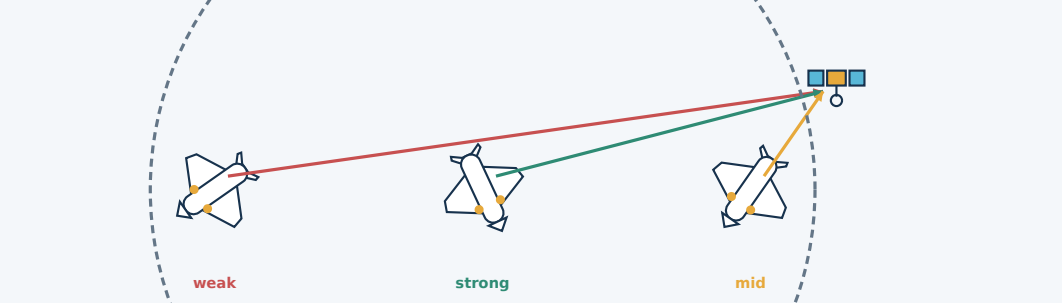


Perspective geometry triad

Toward, broadside and away at low satellite elevation, with the side-mounted arrays called out.

EXPLAIN + MODEL
Opening figure: Immediate intuition

02 GEOMETRY

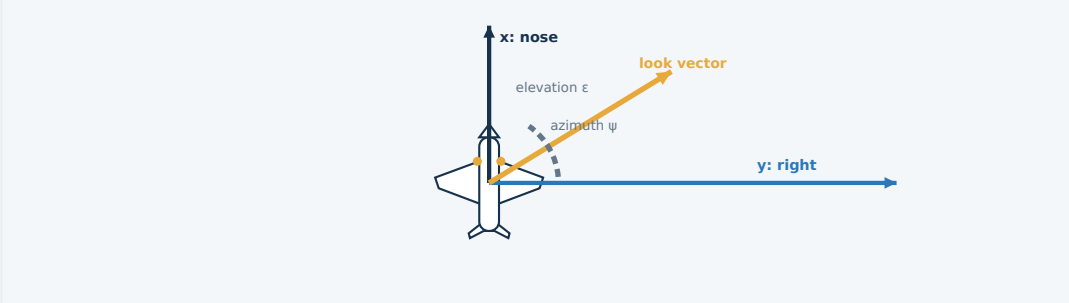


Rotate the aircraft, hold the satellite fixed

Three headings at one ground point isolate why heading—not position alone—changes antenna gain.

EXPLAIN
Opening alternative / animation

03 GEOMETRY

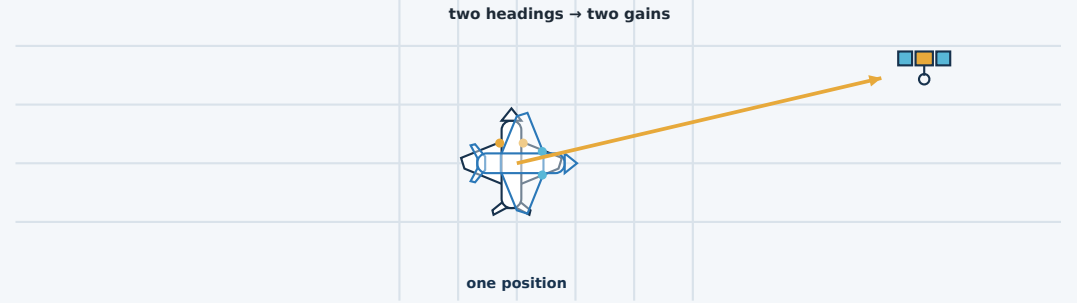


Aircraft-space coordinate frame

Defines relative azimuth and elevation, body axes, attitude and the satellite line-of-sight vector.

EXPLAIN + DEFINITIONS
Technical inset beside equations

04 GEOMETRY

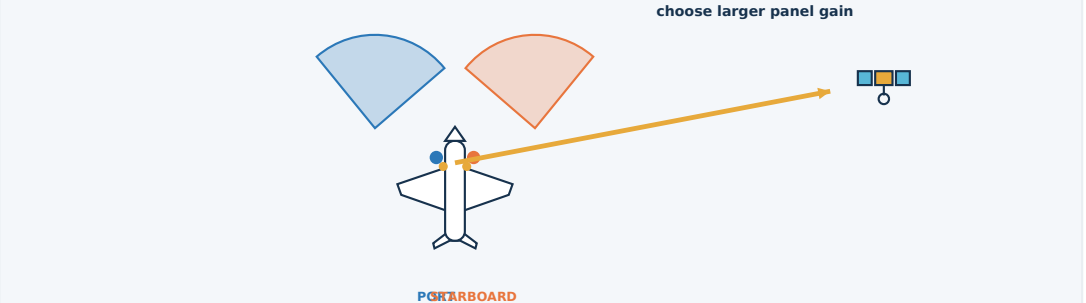


Same map position, different received power

A plan-view counterexample: identical aircraft and satellite positions, but different headings and gains.

EXPLAIN
General-reader bridge

05 GEOMETRY

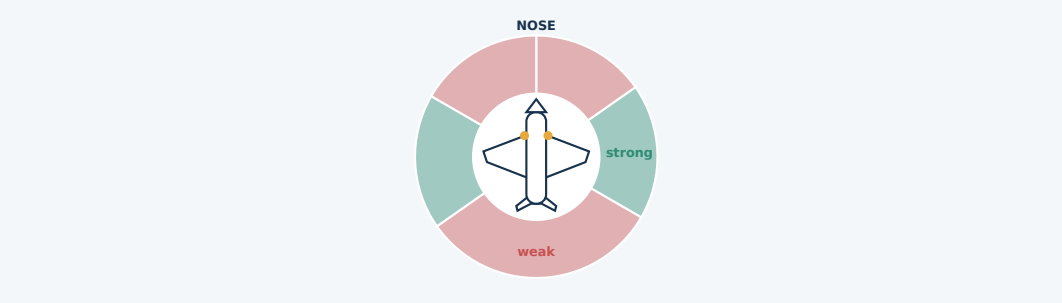


Port/starboard antenna hand-off

Shows the two canted side arrays and which panel supplies the larger gain as the look direction moves.

EXPLAIN + MODEL
Hardware/geometry explanation

06 GEOMETRY

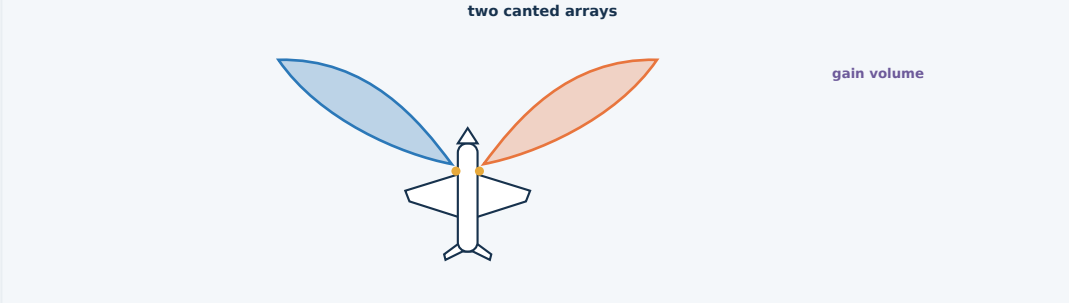


Horizon compass of strong and weak sectors

A top-down coverage ring makes the low-elevation nose/tail weakness visible without a 3-D surface.

MODEL
Compact main-text alternative

07 GAIN

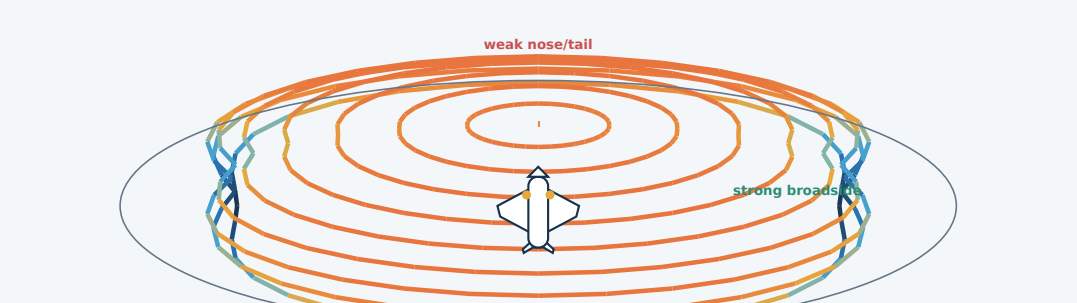


Twin gain lobes mounted on the airframe

An aircraft carries the two directional gain volumes, connecting the side-mounted hardware to the pattern.

EXPLAIN + MODEL
Hero alternative / graphical abstract

08 GAIN

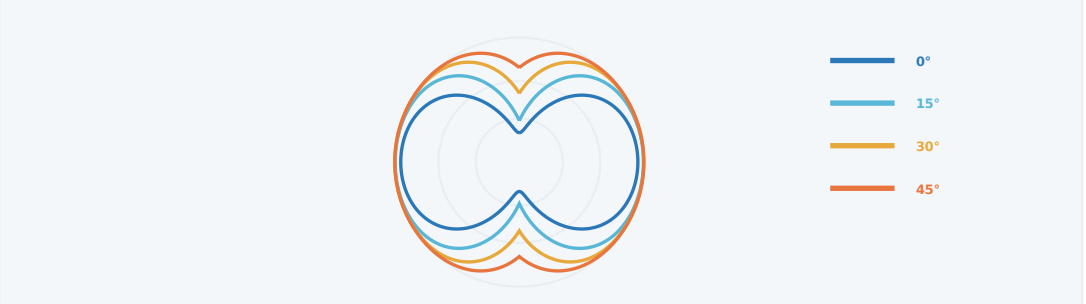


3-D gain hemisphere

A colour-coded upper-hemisphere surface shows broadside maxima and low-elevation nose/tail valleys.

MODEL
Primary gain-pattern figure

09 GAIN

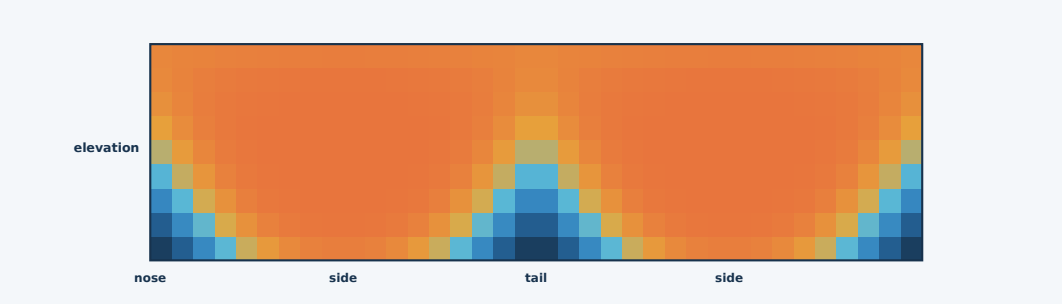


Polar slices at several elevations

Azimuthal gain curves at 0°, 15°, 30° and 45° elevation improve the dissertation's two-slice plot.

MODEL + DATA
Quantitative main or supplement

10 GAIN

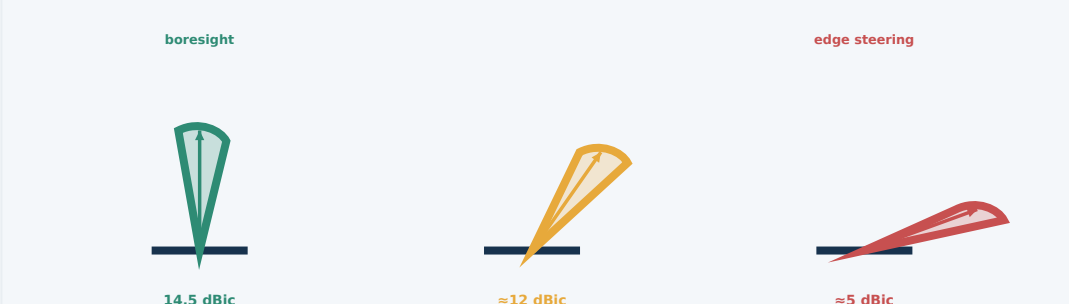


Elevation-azimuth gain heat map

A rectangular aircraft-space map is easiest for readers to query and for particles to interpolate.

MODEL + DATA
Methods figure / model validation

11 GAIN

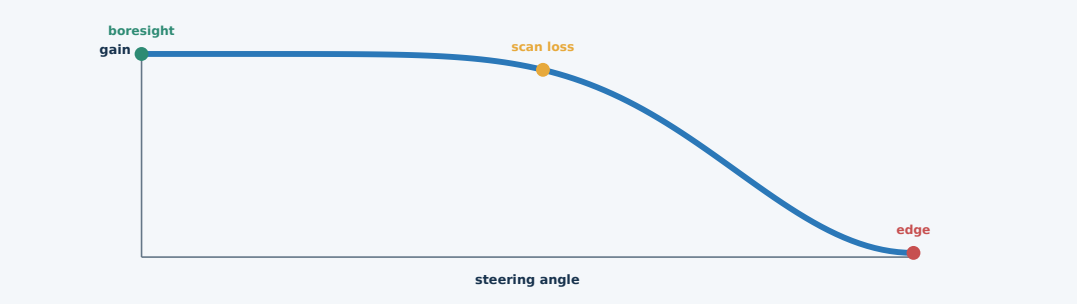


Beam steering and scan loss

A three-step diagram follows the beam from boresight toward the array edge as peak gain falls.

EXPLAIN
Phased-array explainer

12 GAIN

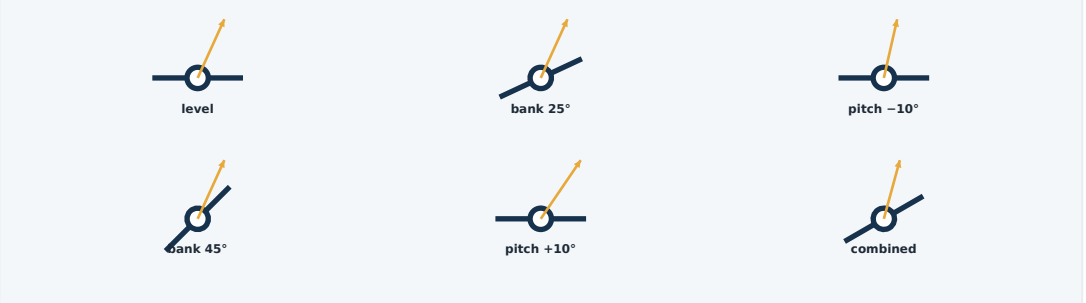


Bore-sight-to-horizon gain profile

A single annotated curve turns the 3-D pattern into the one-dimensional quantity used in the link budget.

MODEL + DATA
Compact quantitative inset

13 ATTITUDE

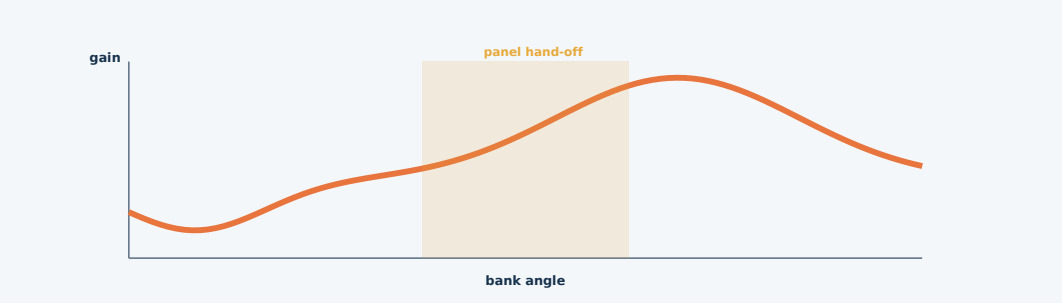


Pitch-bank-yaw coverage matrix

Small multiples show how the same spacecraft direction moves across the gain map under attitude changes.

MODEL
Controlled/unusual-flight discussion

14 ATTITUDE

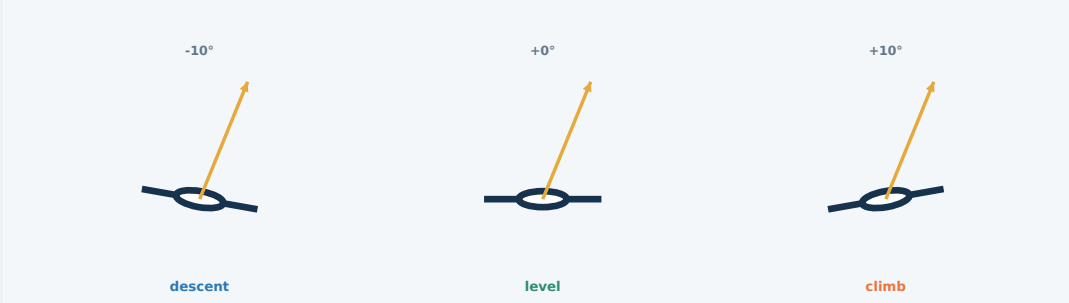


Bank-angle sweep

A continuous sweep shows when a low-elevation spacecraft enters a strong lobe, weak sector or hand-off region.

MODEL
Sensitivity figure

15 ATTITUDE

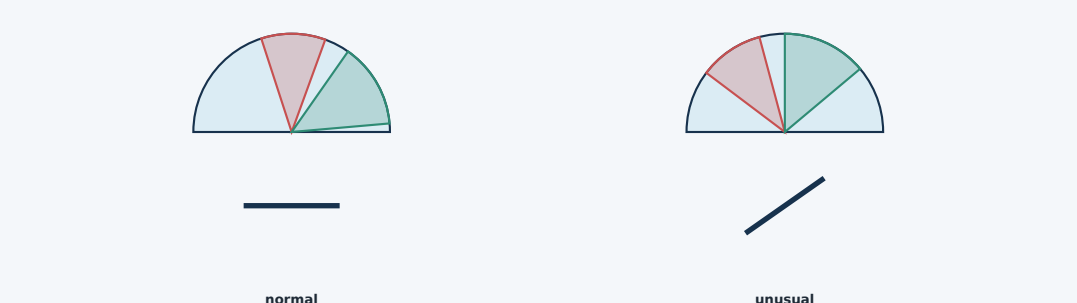


Climb and descent sweep

Contrasts level flight, descent and climb without conflating flight-path angle with aircraft-space elevation.

MODEL
Controlled-flight scenario inset

16 ATTITUDE

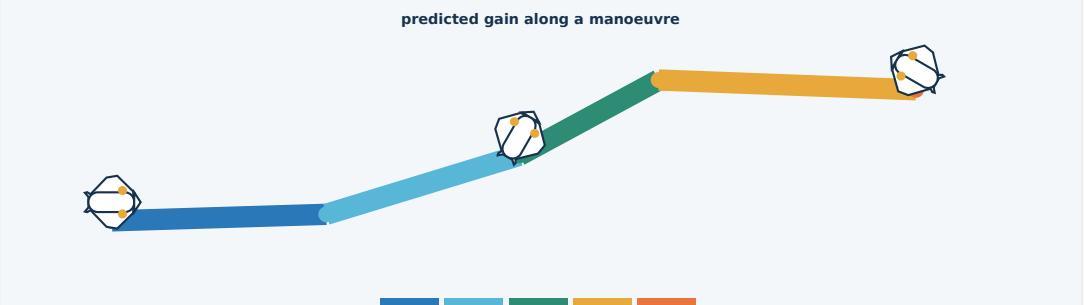


Normal versus unusual attitude sky domes

Two hemispheres compare routine cruise coverage with a steep bank/pitch case, including no-coverage regions.

MODEL
End-of-flight caveat / supplement

17 ATTITUDE

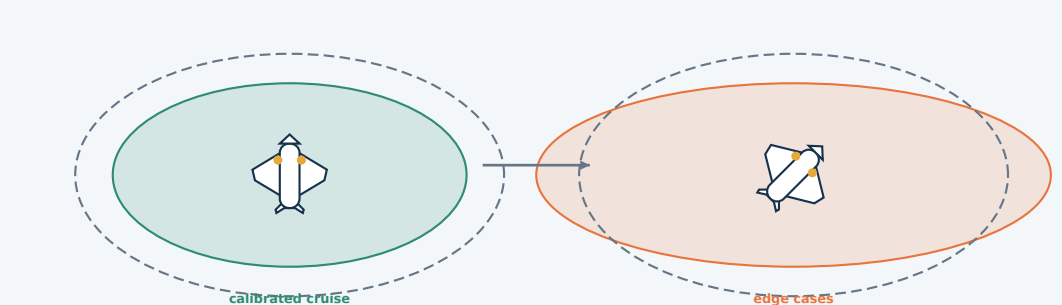


Coverage ribbon through a controlled turn

A route ribbon is coloured by predicted gain while heading and bank evolve through a turn.

MODEL + TRAJECTORY
Trajectory-model integration

18 ATTITUDE



Operational envelope and edge cases

A two-panel map separates well-constrained normal-attitude behaviour from extrapolative unusual attitudes.

MODEL + UNCERTAINTY
Uncertainty disclosure

19 MH371

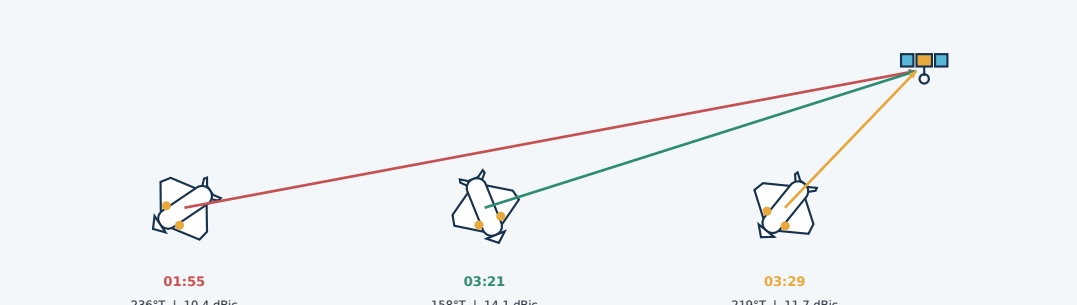


MH371 route as a natural experiment

The route map marks 01:55, 03:21 and 03:29, where heading changes produced large predicted gain changes.

DATA
Primary validation figure

20 MH371

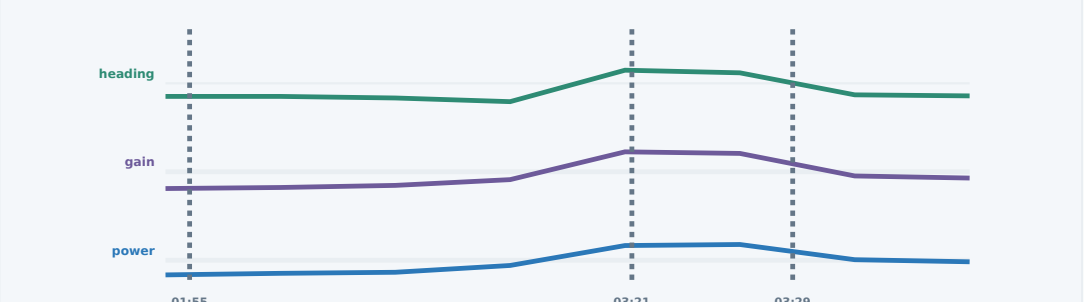


Three matched geometry snapshots

At each phase, a plane, line of sight, heading, predicted gain and path-corrected power are shown together.

DATA + MODEL
General-reader validation figure

21 MH371

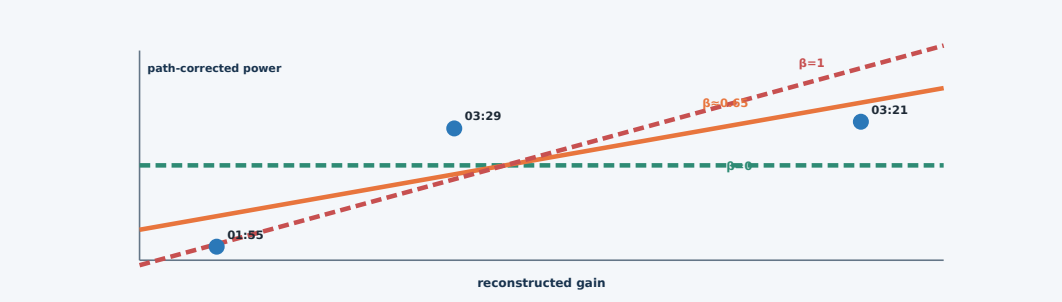


Time-aligned heading, gain and power

Stacked traces show whether received-power changes occur where the geometry model predicts them.

DATA + MODEL
Technical validation figure

22 MH371

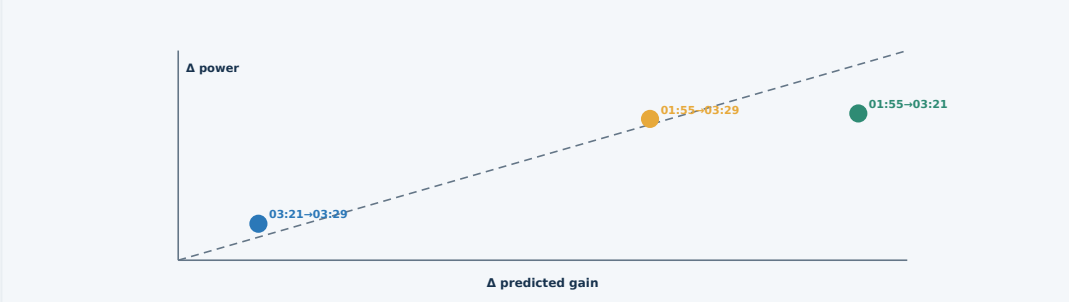


Same-channel counterfactual test

The three phase means are compared with complete, partial and zero gain-precompensation predictions.

DATA + INFERENCE
Core hypothesis-test figure

23 MH371

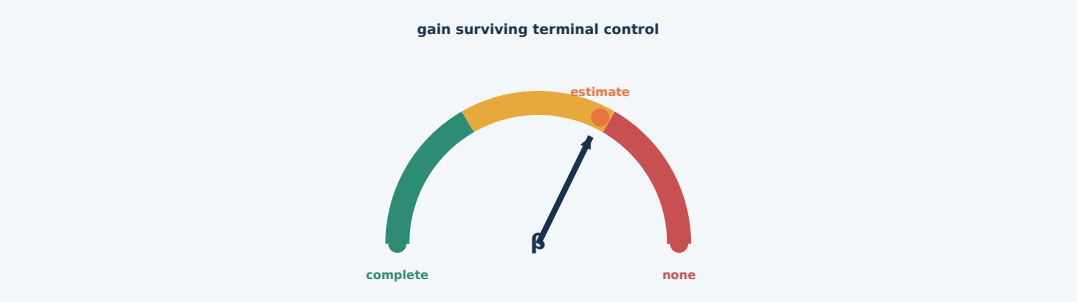


Observed change versus predicted gain change

A change-on-change plot removes the arbitrary power intercept and makes agreement or attenuation visible.

DATA + INFERENCE
Robustness / supplement

24 INFERENCE

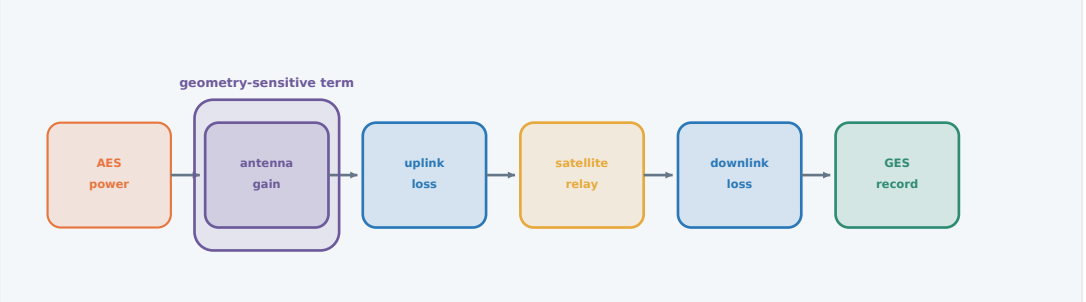


Gain-precompensation coefficient β

A visual continuum links $\beta=0$ (complete compensation), the estimate, and $\beta=1$ (none).

INFERENCE
Hypothesis-test summary

25 INFERENCE

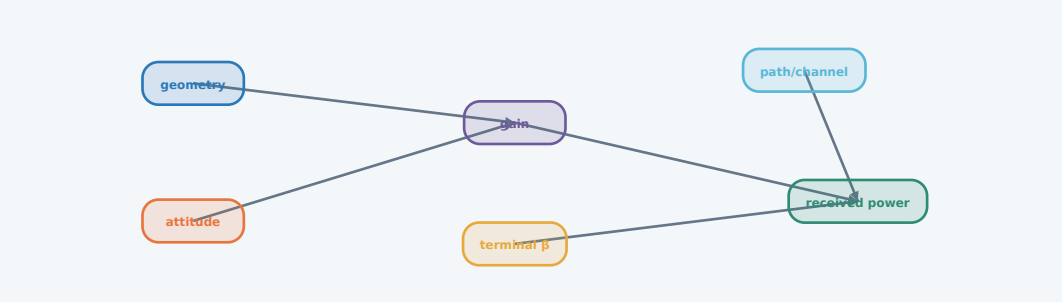


End-to-end received-power chain

A left-to-right link budget separates AES gain, space loss, satellite/ground terms and recorded power.

EXPLAIN + MODEL
Methods overview

26 INFERENCE

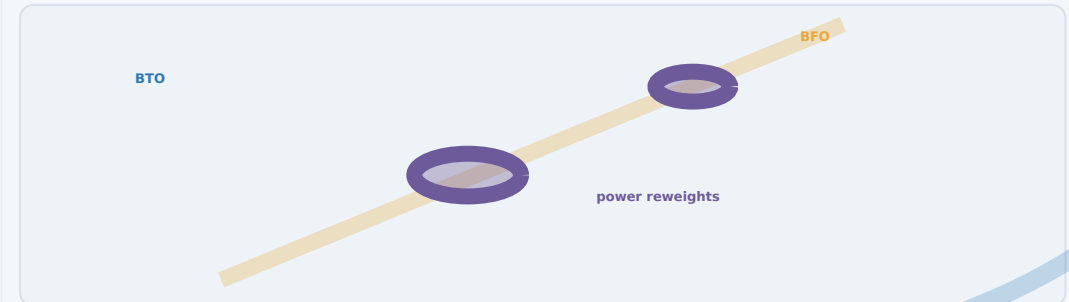


Causal graph and nuisance separation

Geometry drives gain; channel, path, terminal control and slower RF offsets also influence the observation.

INFERENCE + MODEL
Statistical-model architecture

27 INFERENCE

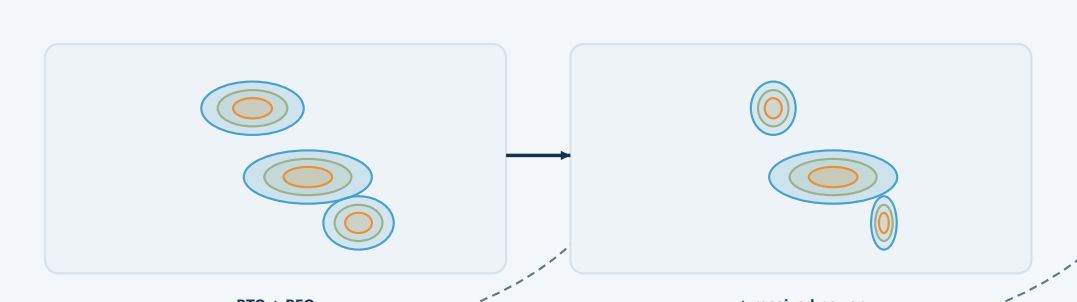


How power intersects the BTO/BFO solution

Layered likelihood ridges show power narrowing—but not independently locating—the satellite-constrained path.

INFERENCE
Integrated-estimator explanation

28 INFERENCE



Posterior before and after received power

Side-by-side 00:11 maps reveal which modes are reweighted when the received-power likelihood is added.

INFERENCE + DATA
Results figure